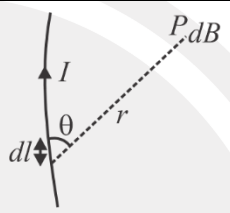
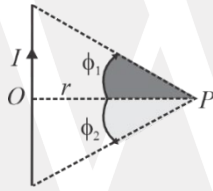
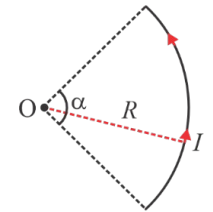
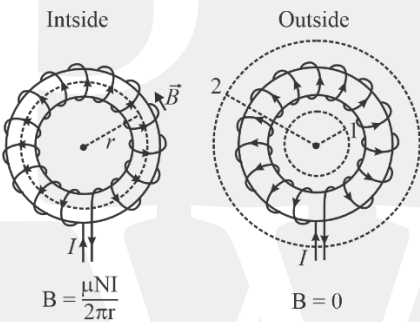
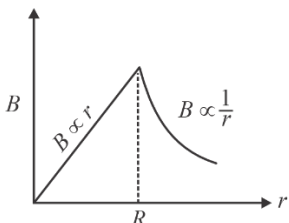
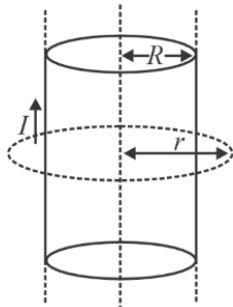


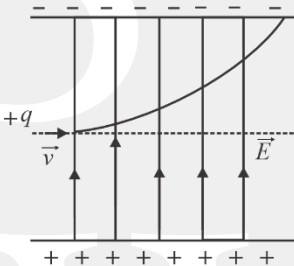
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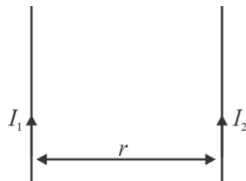
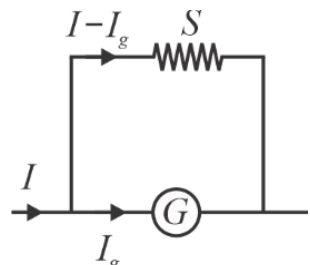
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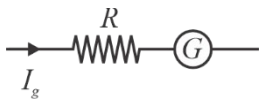
Moving charges and magnetism

Sr. No.	Concept	Formulas	Other information
1.	Biot-savart's law	$dB = \frac{\mu_0 i dl \sin \theta}{4\pi r^2}$ $d\vec{B} = \frac{\mu_0}{4\pi} i \frac{(d\vec{l} \times \vec{r})}{r^3}$  The direction of magnetic field $d\vec{B}$ is that of $(d\vec{l} \times \vec{r})$	dB = magnetic field produced by a current element. dl = length of current element. I = current flowing. μ_0 = permeability of free space.
2.	Magnetic field due to thin wire.	For a finite wire: $B_{net} = \frac{\mu_0 I}{4\pi r} (\sin \phi_1 + \sin \phi_2)$  For infinite length wire: $B = \frac{\mu_0}{4\pi} \cdot \frac{2I}{r}$	I = the amount of current flowing through the conductor. r = the perpendicular distance.
3.	Magnetic field due to circular arc at the centre	$B = \frac{\mu_0}{4\pi} \frac{I\alpha}{R} \quad (\alpha \text{ must be in radian only})$ 	α = arc angle (must be in radian only) R = radius of the arc I = current flowing through wire
4.	Magnetic field due to circular current loop.	(i) At the centre of the loop: $B = \frac{\mu_0 I}{2a}$ (ii) At an axial point at distance r from the centre: $B = \frac{\mu_0 I a^2}{2(r^2 + a^2)^{3/2}}$	a = radius of circular loop. I = current flowing through loop.

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5.	Ampere's circuital law	The line integral of the magnetic field B around any closed circuit: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$	μ_0 = permeability of free space. I = current passing through this closed loop.
6.	Magnetic field of a infinite long solenoid.	(i) $B = \mu_0 n I$ (At a point well inside the solenoid) (ii) $B_{end} = \frac{1}{2} \mu_0 n I$ (At either end of the solenoid)	B = the magnetic field of a straight solenoid. I = current in the solenoid. n = number turns per unit length.
7.	Magnetic field due to toroid.	Magnetic field due to current carrying toroid. $B = \mu_0 n I$ Magnetic field outside the toroid and also in empty space surrounded by toroid, $B = 0$ because net current enclosed is zero. 	I = the current in the windings. n = the number of turns per unit length.
8.	Magnetic field due to current carrying long cylinder.	Outside the cylinder ($r > R$) $B = \frac{\mu_0 I}{2\pi r}$ Inside the cylinder when it is made of a thin metal sheet, $B = 0$ Inside the cylinder when current is uniformly distributed throughout the cross-section of the cylinder ($r < R$) $B = \frac{\mu_0 \mu_r I r}{2\pi R^2}$ 	μ_r = relative permeabilities material of the cylinder. I = current flowing through the cylinder. R = radius of the cylinder. 

Sr. No.	Concept	Formulas	Other information
9.	Force on a charge moving in a magnetic field.	Magnetic force, $F = qvB \sin \theta$ In vector notation, $\vec{F} = q(\vec{v} \times \vec{B})$	q = charge moving with velocity v . θ = angle made with magnetic field B .
10.	Lorentz force	The total electromagnetic force on a charge particle $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$	F = the total force acting on a charge q . E = electric field. B = magnetic field.
11.	Motion of charge inside an electric field.	Electric field set up between the plates is $E = \frac{V}{d}$ The charge q of mass m experiences the electric force, $\vec{F}_e = q\vec{E}$ Acceleration produced in the charge, $\vec{a} = \frac{q\vec{E}}{m}$ 	V = potential difference applied between two parallel plates. d = separation between the plates.
12.	Motion of charge inside a magnetic field.	The radius of the path, $r = \frac{mv}{Bq}$ Time period, $T = \frac{2\pi m}{Bq}$ When charged particle enter magnetic field at any angle except 0° , 90° and 180° , then it follows helical path. The radius of the path, $r = \frac{mv \sin \theta}{Bq}$ Pitch = $T \times v \cos \theta = \frac{2\pi m v \cos \theta}{Bq}$	v = speed of particle B = the applied magnetic field. q = charge of the particle m = the mass of the charged particle. $v \cos \theta$ = pitch of the path.
13.	Force on a current-carrying conductor in a magnetic field.	Magnetic force: $F = I l B \sin \theta$ In vector notation, $\vec{F} = I (\vec{l} \times \vec{B})$	l = length of conductor. I = current flowing through the conductor. B = magnetic field.

Sr. No.	Concept	Formulas	Other information
14.	Force between two parallel infinitely long current-carrying conductors.	$F = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$ The force is attractive if current in both conductors is in same direction and repulsive if current in both conductors is in opposite direction. 	F = the force per unit length between two long parallel conductors. I_1 and I_2 = currents carrying r = distance between two parallel wires.
15.	Torque on current carrying coil in a magnetic field.	$\tau = NIBA \sin \theta = mB \sin \theta$ In vector notation, $\vec{\tau} = \vec{m} \times \vec{B}$ Torque is maximum when the plane of the coil is parallel to the magnetic field ($\theta = 90^\circ$) $\tau_{\max} = NIBA$.	N = the number of turns in the coil $m = NIA$ is magnetic dipole moment. θ = angle which the normal to the plane of the coil makes with the field B.
16.	Magnetic moment and Bohr magneton (μ_B) for a revolving electron	Magnetic moment for an electron; $M = \frac{neh}{4\pi m}$ For, $n = 1$, it is known as Bohr magneton $\mu_B = \frac{eh}{4\pi m} = \frac{1.6 \times 10^{-19} \times 6.6 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31}}$ $= 0.923 \times 10^{-23} \text{ Am}^2$	M = magnetic moment e = charge on electron h = Planck's constant n = principal orbit number m = mass of electron
17.	Moving coil galvanometer.	Restoring torque = Deflecting torque or $k\alpha = NIBA$ or $\alpha = \frac{NBA}{k} \cdot I$ Current sensitivity = $\frac{\alpha}{I} = \frac{NBA}{k}$ Voltage sensitivity = $\frac{\alpha}{V} = \frac{\alpha}{IR} = \frac{NBA}{kR}$	A = area of the coil. N = number of turns. B = the magnetic field. k = restoring torque per unit twist. α = deflection produced in galvanometer.
18.	Conversion of a galvanometer into an ammeter.	$S = \frac{I_g}{I - I_g} \times G$ Total resistance of an ammeter; $R_A = \frac{GS}{G + S}$ 	G = resistance of galvanometer. I_g = the current with which galvanometer gives full-scale deflection. S = shunt (low resistance)

Sr. No.	Concept	Formulas	Other information
19.	Conversion of galvanometer into a voltmeter	The value of R is given by; $R = \frac{V}{I_g} - G$ Total resistance of a voltmeter, $R_V = G + R$ 	G = galvanometer resistance. R = external resistance in series with G.

